Two-Way Tables & Probability

1) After school on Tuesday,

14 Year 9s played football. 12 Year 9s played basketball.
17 Year 10s played football. 11 Year 10s played basketball.

Complete the two-way table with this information

	Football	Basketball	Total
Year 9			
Year 10			
Total			

a) How many students played Basketball?
b) How many students were there in total?
c) If a student was picked at random, what is the probability they play Basketball? (express your answer as a fraction)

2) 60 students were asked if they prefer Maths, Science or English.
Complete the two-way table.

	Maths	Science	English	Total
Year 7	12	8		
Year 8	11		10	35

a) If a student was picked at random, what is:
P(They prefer Maths)?
b) If we picked **only from** Year 7s, what is:
P(They prefer English)?

3) 40 adults & 50 children people were asked how they prefer to keep fit.
Complete the two-way table.

	Running	Gym	Sport	N/A	
Adults	17	10		4	
Children		6	21		
Total	37				

a) From everyone, what is P(They prefer to play sport)?
b) From only those that prefer going to the gym, what is P(Adult)?

4) 100 mixed Year 7, 8 & 9 students were asked how they got to school.

For Year 7s: 12 walked, 9 cycled, 7 took a bus and 2 took a car.
For the 30 Year 8s: 5 walked, 11 cycled & 9 took a bus
15 of the Year 9 students cycled to school.
In total, 33 students walked and 22 took the bus.

Complete a two-way table with this information.

a) What is P(Cycled and are in Year 9)?
b) What is P(Year 7 and walked)?
c) Of those that took a car, what is P(Year 7)?

5) Children under 13, Teenagers, Adults & Pensioners were asked how much money they spend a day on snacks

	0-99p	£1-1.99	£2+	Total
Child (<13)		4		14
Teenager	4	8	2	
Adult	5			
Pensioner		5	4	11
Total	14	22		50

a) Find P(A person who spends £2 or more a day)
b) Find P(A teenager who spends more than £1 per day)
c) From those older than 12, P(A person who spends less than £2 per day)

Two-Way Tables & Probability

1) After school on Tuesday,

14 Year 9s played football. 12 Year 9s played basketball.
17 Year 10s played football. 11 Year 10s played basketball.

Complete the two-way table with this information.

	Football	Basketball	Total
Year 9	14	12	26
Year 10	17	11	28
Total	31	23	54

a) How many students played Basketball? 23
b) How many students were there in total? 54
c) If a student was picked at random, what is the probability they play Basketball? (express your answer as a fraction) $\frac{23}{54}$

2) 60 students were asked if they prefer Maths, Science or English.
Complete the two-way table.

	Maths	Science	English	Total
Year 7	12	8	5	25
Year 8	11	14	10	35
Total	23	22	15	60

a) If a student was picked at random, what is: P(They prefer Maths)? $\frac{23}{60}$
b) If we picked **only from** Year 7s, what is: P(They prefer English)? $\frac{5}{25} = \frac{1}{5}$

3) 40 adults & 50 children people were asked how they prefer to keep fit.
Complete the two-way table.

	Running	Gym	Sport	N/A	Total
Adults	17	10	9	4	40
Children	20	6	21	3	50
Total	37	16	30	7	90

a) From everyone, what is P(They prefer to play sport)? $\frac{1}{3}$
b) From only those that prefer going to the gym, what is P(Adult)? $\frac{10}{16} = \frac{5}{8}$

4) 100 mixed Year 7, 8 & 9 students were asked how they got to school.

For Year 7s: 12 walked, 9 cycled, 7 took a bus and 2 took a car.
For the 30 Year 8s: 5 walked, 11 cycled & 9 took a bus
15 of the Year 9 students cycled to school.
In total, 33 students walked and 22 took the bus.

Complete a two-way table with this information.

	Walk	Cycle	Bus	Car	Total
Year 7	12	9	7	2	30
Year 8	5	11	9	5	30
Year 9	16	15	6	3	40
Total	33	35	22	10	100

a) What is P(Cycled and are in Year 9)? $\frac{15}{100} = \frac{3}{20}$
b) What is P(Year 7 and walked)? $\frac{12}{100} = \frac{3}{25}$
c) Of those that took a car, what is P(Year 7)? $\frac{2}{10} = \frac{1}{5}$

5) Children under 13, Teenagers, Adults & Pensioners were asked how much money they spend a day on snacks.

	0-99p	£1-1.99	£2+	Total
Child (<13)	3	4	7	14
Teenager	4	8	2	14
Adult	5	5	1	11
Pensioner	2	5	4	11
Total	14	22	14	50

a) Find P(A person who spends £2 or more a day) $\frac{14}{50} = \frac{7}{25}$
b) Find P(A teenager who spends more than £1 per day) $\frac{12}{14} = \frac{6}{7}$
c) From those older than 12, P(A person who spends less than £2 per day) $\frac{29}{36}$